

Hybrid Energy Based and Normalizing Flow Models for Trustworthy LLM Output Evaluation

Group name : Infinity Zero
Sayan Ghosh (B2530087)
Pritam Kayal (B2530096)

April 27, 2026

Abstract

Large Language Models (LLMs) generate fluent text but often suffer from hallucinations, inconsistency, and lack of reliability. This project proposes a hybrid framework combining Energy-Based Models (EBMs) and Normalizing Flow Models (NFMs) to evaluate the trustworthiness of LLM-generated outputs. EBMs are used to assign unnormalized confidence scores to detect anomalous or low-quality responses, while Normalizing Flows provide exact likelihood estimation for uncertainty quantification. The hybrid approach enables both interpretability and probabilistic evaluation of outputs. Experiments are conducted on LLM-generated responses across factual and reasoning tasks. The proposed system improves robustness, uncertainty awareness, and reliability in LLM evaluation.

1 Introduction

- What is the project?

This project focuses on evaluating the trustworthiness of outputs generated by Large Language Models (LLMs). Instead of judging responses only by fluency or similarity-based metrics, the project proposes a hybrid evaluation framework that combines Energy-Based Models (EBMs) and Normalizing Flow Models (NFMs). The system assigns a quantitative trust score to each LLM-generated output, indicating how reliable, consistent, and uncertainty-aware the response is.

- Why is this project needed?

Although LLMs generate human-like text, they often produce hallucinations, overconfident incorrect answers, and logically inconsistent responses. Existing evaluation metrics fail to capture uncertainty and cannot reliably detect unreliable outputs. This creates risks when LLMs are used in high-stakes applications such as education, healthcare, and decision support systems. Therefore, a probabilistic and interpretable evaluation mechanism is required to assess not just what the model says, but how trustworthy the output is.

- How is the problem addressed?

The project uses a hybrid modeling approach :

- Energy-Based Models (EBMs) assign energy scores to LLM outputs, where low energy represents trustworthy responses and high energy indicates anomalous or unreliable outputs.
- Normalizing Flow Models (NFMs) model the probability distribution of high-quality responses and provide exact likelihood-based uncertainty estimation.
- These two scores are combined to compute a final trustworthiness score, enabling reliable detection of hallucinated or low-confidence LLM outputs.

2 Literature Review

1. Hallucination in LLMs

- LLMs can produce high-confidence wrong answers
- Problem : No proper uncertainty estimation

2. Energy-Based Models

- Learn energy function:
 - Low energy $\rightarrow$ correct
 - High energy $\rightarrow$ incorrect
- Used in anomaly detection

3. Normalizing Flows

- Learn probability distribution
- Provide exact likelihood
- Used for uncertainty modeling

Limitations of Existing Methods

Method	Problem
Accuracy	Cannot detect hallucination
Softmax confidence	Overconfident
EBM only	No probability
NF only	No discriminative scoring

Our Contribution

combine :

- Discriminative (EBM)
- Generative (NF)

Result : More reliable + uncertainty-aware system

3 Proposed methodology

Overall Pipeline

- Straight pipeline : Input $\rightarrow$ BERT $\rightarrow$ Feature
- Split into two branches:
 - Left $\rightarrow$ EBM $\rightarrow$ Energy
 - Right $\rightarrow$ NF $\rightarrow$ Likelihood
- Merge $\rightarrow$ Hybrid Score
- Final output $\rightarrow$ Trust / Hallucination

3.1 Dataset

- Dataset: TruthfulQA.csv
 - Total samples: 790
 - Total features (columns): 8
- Contains :
 - Question
 - Correct Answer
 - Incorrect Answers

3.2 Data Preparation

- Positive sample : Question + Correct Answer $\rightarrow$ Label 0
- Negative sample : Question + Incorrect Answer $\rightarrow$ Label 1

3.3 Feature Extraction

- Model : BERT (bert-base-uncased)
- Input : [CLS] Question + Answer
- Output : CLS embedding $\rightarrow$ feature vector

3.4 Energy-Based Model (EBM)

Idea

Assign energy to each input.

- Low Energy $\rightarrow$ Trustworthy
- High Energy $\rightarrow$ Hallucinated

Formula : $E(x) = f_{\theta}(x)$

Training :

Contrastive Learning : $L = E(x_{\text{positive}}) - E(x_{\text{negative}})$

3.5 Normalizing Flow Model (NF)

Idea :

Model probability distribution of correct answers.

Transformation : $z = f(x)$

Likelihood : $\log p(x) = \log p(z) + \log \left| \det \frac{\partial f}{\partial x} \right|$

Interpretation :

- High likelihood $\rightarrow$ reliable answer
- Low likelihood $\rightarrow$ anomaly

3.6 Hybrid Trust Score

Final score:

$$\text{Trust Score} = \alpha \cdot (-E(x)) + (1 - \alpha) \cdot \log p(x)$$

α = weight

3.7 Fusion Trust Score

$$\text{LLR} = \log p_h(z) - \log p_t(z)$$

$$\text{Trust Score} = \sigma(E, \log p_t, \log p_h, \text{LLR}, \dots)$$

4 Experimental Results & Detailed Analysis

4.1 Model Behavior Overview

Our system evaluates whether LLM outputs are truthful vs hallucinated using three approaches:

- EBM (Energy-Based Model) $\rightarrow$ discriminative confidence scoring
- NF (Normalizing Flow: RealNVP, MAF) $\rightarrow$ density estimation
- Hybrid / Fusion $\rightarrow$ combines both

4.2 Individual Model Performance Analysis

1. EBM Only

- **Accuracy** : 74.25%
- **F1** : 0.818
- **AUROC** : 0.861
- **Balanced Accuracy** : 0.776
- **Precision** : 0.94
- **Confusion Matrix** :

$$\begin{bmatrix} 99 & 20 \\ 136 & 351 \end{bmatrix}$$

Interpretation

- Very high precision (0.94) → when it says “hallucinated”, it is usually correct
- Lower recall → misses some hallucinations
- Good balance between truthful & hallucinated detection

2. NF Only (RealNVP)

- **Accuracy** : 75.24%
- **F1** : 0.833
- **AUROC** : 0.777
- **Balanced Accuracy** : 0.731
- **Precision** : 0.911
- **Confusion Matrix** :

$$\begin{bmatrix} 83 & 36 \\ 114 & 373 \end{bmatrix}$$

Interpretation

- Better recall than EBM → detects more hallucinations
- Worse AUROC → weaker overall separation
- Truthful accuracy drops (0.69)

NF is **good at detecting hallucinations, but poor at modeling truthful samples.**

3. NF Only (MAF)

- **Accuracy** : 72.77%
- **F1** : 0.818
- **AUROC** : 0.740
- **Balanced Accuracy** : 0.668
- **Precision** : 0.879
- **Confusion Matrix** :

$$\begin{bmatrix} 68 & 51 \\ 114 & 373 \end{bmatrix}$$

Interpretation

- Better Worst among all
- Very poor truthful detection (57%)

MAF alone is **not reliable for classification.**

4.3 Hybrid Model Analysis

- Hybrid (RealNVP)
 - **Accuracy** : 74.25%
 - **F1** : 0.8181
 - **AUROC** : 0.8613
 - **Precision** : 0.946
 - **Balanced Accuracy** : 0.776
 - **Best Alpha** : 1.0
 - **Confusion Matrix** :

$$\begin{bmatrix} 99 & 20 \\ 136 & 351 \end{bmatrix}$$

Interpretation

- NF contribution **zero**
- Model fully depends on EBM

- Hybrid (MAF)
 - **Accuracy** : 74.25%
 - **F1** : 0.819
 - **AUROC** : 0.862
 - **Precision** : 0.943
 - **Balanced Accuracy** : 0.773
 - **Best Alpha** : 0.65
 - **Confusion Matrix** :

$$\begin{bmatrix} 98 & 21 \\ 135 & 352 \end{bmatrix}$$

Interpretation

- Slight improvement over EBM
- **EBM + NF (MAF)** slightly improves representation.

4.4 Fusion Model (BEST PERFORMANCE)

1. Fusion (RealNVP)

- **Accuracy** : 82.84%
- **F1** : 0.887
- **AUROC** : 0.885
- **Precision** : 0.942
- **Balanced Accuracy** : 0.814
- **Confusion Matrix** :

$$\begin{bmatrix} 94 & 25 \\ 79 & 408 \end{bmatrix}$$

2. Fusion (MAF)

- **Accuracy** : 80.36%
- **F1** : 0.868
- **AUROC** : 0.889
- **Precision** : 0.946
- **Balanced Accuracy** : 0.808
- **Confusion Matrix** :

$$\begin{bmatrix} 97 & 22 \\ 97 & 390 \end{bmatrix}$$

4.5 BEST MODEL (Final Conclusion)

Fusion (RealNVP) is the BEST model Why?

1. **Highest Overall Accuracy** : 82.84% (highest among all)
2. **Best F1 Score** : 0.887 $\rightarrow$ best balance of precision & recall
3. **Strong AUROC** : 0.885 $\rightarrow$ best discrimination capability
4. **Balanced Performance**
 - Truthful accuracy: 0.79
 - Hallucinated accuracy: 0.83

4.6 Final Report Statement

The objective was to evaluate whether integrating discriminative and generative approaches could improve classification performance on an imbalanced dataset. The experimental results demonstrate that: The Energy-Based Model (EBM) alone provides strong performance, effectively separating truthful and hallucinated responses using learned energy scores. Normalizing Flow models (RealNVP and MAF), when used independently, show relatively weaker performance, indicating that density estimation alone is insufficient for accurate hallucination detection.

The Hybrid approach provides moderate improvement, particularly when meaningful contribution from the NF component is present (as seen in the MAF-based hybrid model). The Fusion models achieve the best overall performance by combining EBM and NF outputs, demonstrating the advantage of integrating discriminative and generative signals. Among all models, the Fusion model with MAF achieved the highest performance in terms of AUROC and overall classification quality, while the Fusion model with RealNVP performed best in terms of balanced accuracy, making it more robust to dataset imbalance.

5 Summary

This project successfully developed a hybrid framework combining Energy-Based Models and Normalizing Flows for evaluating LLM trustworthiness.

Key Achievements :

- Detect hallucinated outputs
- Provide uncertainty estimation
- Combine discriminative + generative modeling
- Achieved strong performance (AUROC 0.86)

6 Final Conclusion

Overall, this work highlights that: Discriminative models (EBM) are more effective for direct classification tasks. Generative models (NF) capture useful structural information but are insufficient alone. Combining both approaches leads to more robust and reliable hallucination detection.

7 Future Work

Although the proposed hybrid framework combining Energy-Based Models (EBM) and Normalizing Flows (NF) shows promising results, several improvements can be explored to further enhance performance and robustness. Although the proposed hybrid framework combining Energy-Based Models (EBM) and Normalizing Flows (NF) shows promising results, several improvements can be explored to further enhance performance and robustness.

1. Improving Class Imbalance Handling

2. Stronger Energy-Based Modeling
3. More Expressive Normalizing Flows
4. End-to-End Joint Training
5. Better Fusion Strategies
6. Domain Generalization and Larger Datasets
7. Real-Time Deployment and Optimization

8 References

1. Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville.
2. **Trust Me, I'm Wrong: High-Certainty Hallucinations in LLMs**
Shows that LLMs can hallucinate even when uncertain estimation fails - important for trust research.
3. **A Tutorial on Energy-Based Learning (LeCun et al., MIT Press)**
Seminal EBM tutorial foundational for understanding model formulation and training.
4. **Normalizing Flows: An Introduction and Review of Current Methods**
Survey on flow construction, trade-offs, and design - helps frame why flows are useful for density estimation.
5. **Dataset Source:**
https://huggingface.co/datasets/truthfulqa/truthful_qa;TruthfulQA.csv
6. Google ; ChatGpt .